

Chapter-1 : Introduction to Optimization

[Ref: S.S. Rao Book,
Klarbring Book,
IISc lectures.]

① Introduction:

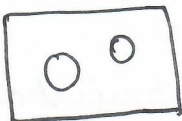
- "Optimization is the act of obtaining the best result under given circumstances".
- "Optimization is achieving the best with the available resources while satisfying the constraints".
- "Mathematical Programming" $\leftrightarrow$ "Optimization" $\leftrightarrow$ "Operations Research".
- No single method for solving all optimization problems efficiently.
- Structural Optimization: "making an assemblage of materials sustain loads in the best way".
- Examples:
 - Design of civil engineering structures (frames, bridges, towers etc.) for minimum cost
 - Minimum weight design of structures for earthquake or wind.
 - Optimizing a structure for stiffness, strength, weight, cost etc.
- Stiff structure design:

What is the "Optimal" structure that could transfer the load to the support?



"Optimal" could be defined in several ways - least weight, ~~or~~ most aesthetic, etc.

- Difference between "Topology", "Shape" and "Size" Optimization.



Topology: How many holes? Deals with 'connectivity' of the material (e.g. no. of holes)



Shape: What is the shape? What should the shape of the holes be? (In these two cases we fix no. of holes).



Size: What are the sizes? (of the holes).

eg. "Stiff structure design" can be obtained by optimizing any of the above three approaches.

② Formal structure of an optimization formulation:

- Statement of an optimization problem (general):

Minimize (objective function)
(optimization variables)
Subject to
(constraints)
(Limits on variables)

In mathematical terms:

$$\min_{x_1, x_2} f(x_1, x_2)$$

$$\text{s.t. } h(x_1, x_2) = 0 \quad \leftarrow \text{equality constraint}$$

$$g(x_1, x_2) \leq 0 \quad \leftarrow \text{Inequality constraint}$$

$$\text{lower} \leftarrow x_1^l \leq x_1 \leq x_1^u \rightarrow \text{upper}$$

$$x_2^l \leq x_2 \leq x_2^u$$

Key words:

- Objective function: The function that has to be optimized
e.g: $f(x_1, x_2)$ (i.e either minimized or maximized)
- Optimization variables (or) Design Variables: (or Decision Variables)

The variables at which the objective function is evaluated. Each variable could be limited in terms of the values that it can take. The goal of the optimization problem is to find the choice of these variables (x_1, x_2) for which the criterion is satisfied (e.g min of $f(x_1, x_2)$) while satisfying all the constraints.

- Constraints: The equations that restrict the choice of design variables that can be considered for optimization.
(e.g: $h(x_1, x_2) = 0$, $g(x_1, x_2) \leq 0$, $x_i^l \leq x_i \leq x_i^u$ etc.)

Can be written similar to $g(x_1, x_2) \leq 0$

$$\text{i.e. } x_i^l - x_i \leq 0$$

$$x_i - x_i^u \leq 0$$

- * "Feasible space": The space (domain) of optimization variables where all constraints are satisfied is called the feasible space.

- * Global vs local optimum:

x = local
o = global



- Statement of a structural optimization problem:

$$\min_{x_i} f(x_i, u) \quad \text{Here } x_i \rightarrow \text{Design variable}$$

$$u \rightarrow \text{state variable}$$

$$\text{s.t. } p(x_i, u) = 0 \rightarrow \text{state equation (e.g. } K(x_i)u = F(x_i) \text{)}$$

$$\left. \begin{array}{l} h(x_i) = 0 \\ g(x_i) \leq 0 \end{array} \right\} \rightarrow \text{could be constraints on weight, cost, size, of the structure strength, stiffness etc.}$$

$$x_i^l \leq x_i \leq x_i^u$$

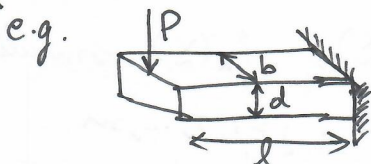
Data: provide material properties, loads etc.

③ Classification of Optimization Problems: (only some relevant classifications are given below)

- classification based on existence of constraints - Constrained vs. Unconstrained optimization.
- classification based on the nature of Design Variables (DVs) -
 - Discrete Variables: ⊕ Binary (only 0 or 1 are DVs), Integers,
 - Continuous → (e.g. ⊕ Discrete sets (e.g. Steel Rebar sizes. Only specific sizes are available)).
 - Deterministic vs. Stochastic:
 - DVs are Random Variables. (probabilistic or non-deterministic).
 - ↳ No uncertainty

- Finite variables or Functions

e.g. $x_1, x_2, \dots$



$$\min_{b, d} \text{Vol} = \rho l b d$$

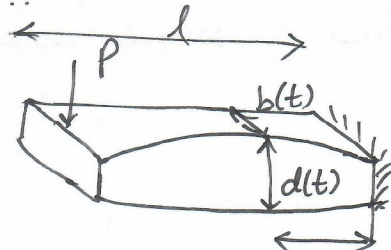
$$\text{s.t. } \delta_{tip}(b, d) \leq \delta_{max}$$

$$b \geq 0$$

$$d \geq 0$$

$x_1(t), x_2(t), \dots$

→ leads to Calculus of Variations.



$$\min_{b(t), d(t)} \text{vol} = \rho \int_0^l b(t) d(t) dt$$

$$\text{s.t. } \delta_{tip}[b(t), d(t)] \leq \delta_{max}, 0 \leq t \leq l$$

$$b(t) \geq 0, 0 \leq t \leq l$$

$$d(t) \geq 0, 0 \leq t \leq l$$

Functional
function of a function.

• Classification based on nature of equations:

- Linear Programming (LP): Both objective function & constraints are linear
- Quadratic Programming: $\downarrow$ Objective " is quadratic & " " linear
- Nonlinear Programming (NLP): Both objective & constraints are nonlinear.

Linear function $\Rightarrow$ ^{Polynomial of degree 1 or less.} ~~$f(x) = a_1x + a_2x + \dots + a_nx$~~ , e.g. $f(x) = 2x + 5$

Nonlinear function: e.g. $f(x) = x^2 + 4$, $f(x) = e^x$ etc.

- Geometric Programming: Objective and constraints are POSYNOIALS.

e.g. of posynomial: $f(x_1, x_2, x_3) = 2.7x_1^2x_2^{-1/3}x_3^{0.7} + 2x_1^{-4}x_3^{2/5}$, $x_i > 0$

Note that the exponents in posynomials are real numbers (+ve or -ve) (unlike polynomials, where they are non-negative integers), and the variables and coefficients must be positive real numbers.

- Convex optimization: Objective function and the feasible space are convex.
- Non-smooth optimization: Objective function or the constraints, or both, are not differentiable. (will see later)

• Classification based on number of objective functions:

- Single objective programming
- Multi-objective programming, let $\underline{x} = [x_1, x_2, \dots, x_n]^T$ are DVs.
Find $\underline{x}$ which minimizes $f_1(\underline{x}), f_2(\underline{x}), \dots, f_k(\underline{x})$ simultaneously
s.t. $g_j(\underline{x}) \leq 0, j = 1, 2, \dots, m$

④ Examples:

Problem(i) (Example 1.1 in S.S. Rao Book)

Design a uniform column of tubular ^{circular} cross-section, with hinge joints at both ends, to carry a compressive load $P = 2500 \text{ kg (force)}$ for minimum cost. Column length = 250 cm.

Material properties: $\sigma_y = 500 \text{ kgf/cm}^2$, $E = 0.85 \times 10^6 \text{ kgf/cm}^2$, $\rho = 0.0025 \text{ kg/cm}^3$

Constraints: • Stress induced in the column should be less than buckling stress as well as the yield stress.

• Mean diameter of column is restricted to be between 2 & 14 cm

• Thicknesses outside 0.2 to 0.8 cm are not available.

Cost of column includes material and construction costs and can be taken as $5W + 2d$, where W is the weight in kilograms-force and d is the mean diameter of the column in centimeters.

Use the "Graphical Optimization" procedure.

Solution:

Design variables (DV's) are mean diameter (d) & tube thickness (t)

$$\underline{x} = \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} d \\ t \end{Bmatrix}$$

Objective function is $f(\underline{x}) = 5W + 2d$

$$= 5\rho l A + 2d$$

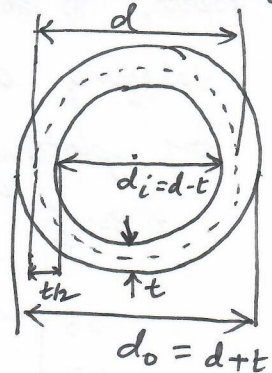
$$= 5\rho l \pi d t + 2d$$

$$= 9.82 x_1 x_2 + 2x_1$$

We need to write constraint equations in terms of "Stress induced", "buckling stress" and "yield stress".

$$\text{Stress induced} = \sigma_i = \frac{P}{A} = \frac{P}{\pi d t} = \frac{2500 (\text{kgf})}{\pi x_1 x_2}$$

$$\text{Buckling stress} = \sigma_b = \frac{P_{cr}}{A} = \frac{\pi^2 E I}{l^2} \cdot \frac{1}{\pi d t}$$



$$\begin{aligned} A &= \frac{\pi}{4} (d_o^2 - d_i^2) \\ &= \frac{\pi}{4} [(d_o + d_i)(d_o - d_i)] \\ &= \frac{\pi}{4} [2d \cdot 2t] = \pi d t \end{aligned}$$

(x_1, x_2 are in cms).

where $I = \frac{\pi}{64} (d_o^4 - d_i^4) = \frac{\pi}{64} (d_o^2 + d_i^2)(d_o^2 - d_i^2)$ Use $d_o = d+t$
 $d_i = d-t$
 $= \frac{\pi}{8} d \cdot t (d^2 + t^2)$
 $= \frac{\pi}{8} x_1 x_2 (x_1^2 + x_2^2) \Rightarrow \sigma_b = \frac{\pi^2 E}{l^2} \cdot \frac{\pi x_1 x_2 (x_1^2 + x_2^2)}{8} \cdot \frac{1}{\pi d t}$

Constraints: (Note $x_1 = d$, $x_2 = t$)

$$g_1(\underline{x}) = \frac{2500}{\pi x_1 x_2} - 500 \leq 0 \quad (\because \sigma_i \leq \sigma_y)$$

$$g_2(\underline{x}) = \frac{2500}{\pi x_1 x_2} - \frac{\pi^2 (0.85 \times 10^6) (x_1^2 + x_2^2)}{8 (250)^2} \leq 0 \quad (\because \sigma_{bi} \leq \sigma_b)$$

Also, $2 \leq d \leq 14$, $0.2 \leq t \leq 0.8$

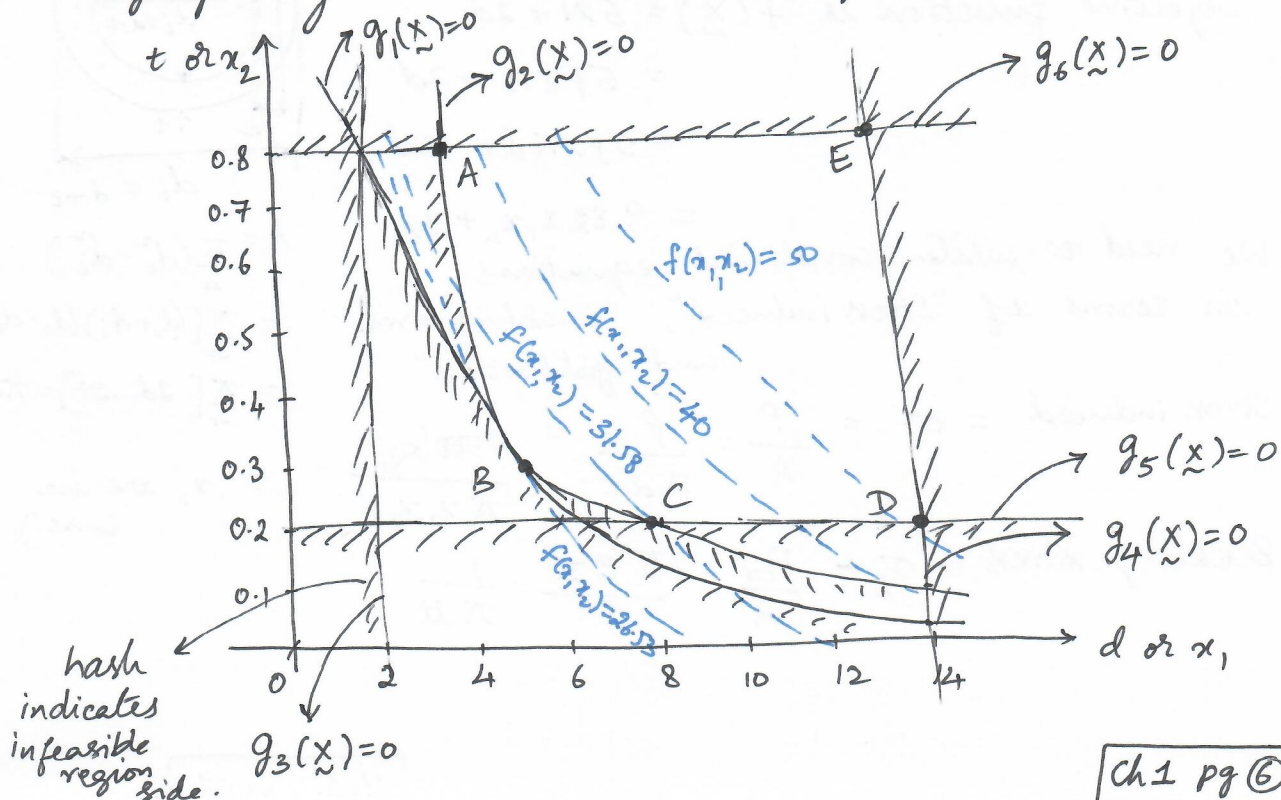
$$\Rightarrow g_3(\underline{x}) = -x_1 + 2 \leq 0$$

$$g_4(\underline{x}) = x_1 - 14 \leq 0$$

$$g_5(\underline{x}) = -x_2 + 0.2 \leq 0$$

$$g_6(\underline{x}) = x_2 - 0.8 \leq 0$$

Now, to solve the problem graphically, consider x_1 & x_2 varying about two axes, and find the "feasible region" by plotting all the constraint equations.



- To find the feasible region, first plot all $g_i(\underline{x})=0$ equations.
 [Then find out which side is feasible by substituting $(x_1, x_2) = (0, 0)$ in $g_i(\underline{x})$, if it satisfies the inequality then the hashed region (infeasible) does not include origin.]
- Next, for various points in the feasible region (ABCDEA), evaluate the value of $f(x_1, x_2)$ objective function and plot its contours to systematically find the (x_1^*, x_2^*) where the minimum $f(x_1^*, x_2^*)$ occurs.
- We find the minimum occurs when $f(x_1, x_2) = 26.53$ (curve) (which passes through the point B). Any other contour of $f(x_1, x_2)$ with lower value than this, does not pass through the feasible region.
- Therefore $(x_1^*, x_2^*) = (5.44, 0.293) = \text{pt B}$ is the optimum pt.
- ⊗ Hence for $d = 5.44 \text{ cm}$, $t = 0.293 \text{ cm}$, the column will entail minimum cost, while ensuring that the induced stress due to the given load is below the yield and buckling stresses.
- * The above problem is a "Minimization of function of several variables subject to constraints".

- * EXERCISE: • Read the "help" section for "fmincon" in MATLAB
- and solve the above problem in MATLAB using "fmincon"
- Plot the 3D plot for $f(x_1, x_2)$ using MATLAB (Surface plot) function.
- Make the 2D projection (contour) plot of $f(x_1, x_2)$ using MATLAB and identify the optimum point obtained.

Problem(ii): A scaffolding system consists of three beams and six ropes. Each of the ropes A and B can carry a load of W_1 , each of the middle ropes C & D can carry a load of W_2 , and each of the bottom ropes E & F can carry a load of W_3 . If the loads acting on beams 1, 2, and 3 are x_1, x_2 , and x_3 , respectively, formulate the problem of finding the maximum load ($x_1 + x_2 + x_3$) that can be supported by the system. Assume that the weights of the beams 1, 2, and 3 are w_1, w_2 , and w_3 , respectively, and the weights of the ropes are negligible.

Solution: let T_i denote the tension in rope i , $i \in \{A, B, C, D, E, F\}$. The rope tensions can be calculated through force & moment equilibrium equations.

For beam 3: $T_E + T_F = x_3 + w_3$

$$x_3(3l) + w_3(2l) - T_F(4l) = 0$$

For beam 2: $T_C + T_D - T_E = x_2 + w_2$

$$x_2(l) + w_2(l) + T_E(l) - T_D(2l) = 0$$

For beam 1: $T_A + T_B - T_C - T_D - T_F = x_1 + w_1$

$$x_1(3l) + w_1\left(\frac{9l}{2}\right) - T_B(9l) + T_C(2l) + T_D(4l) + T_F(7l) = 0$$

Solving these equations, we get $T_F = \frac{3}{4}x_3 + \frac{1}{2}w_3$

$$T_E = \frac{1}{4}x_3 + \frac{1}{2}w_3$$

$$T_D = \frac{1}{2}x_2 + \frac{1}{8}x_3 + \frac{1}{2}w_2 + \frac{1}{4}w_3$$

$$T_C = \frac{1}{2}x_2 + \frac{1}{8}x_3 + \frac{1}{2}w_2 + \frac{1}{4}w_3$$

$$T_B = \frac{1}{3}x_1 + \frac{1}{3}x_2 + \frac{2}{3}x_3 + \frac{1}{2}w_1 + \frac{1}{3}w_2 + \frac{5}{9}w_3$$

$$T_A = \frac{2}{3}x_1 + \frac{2}{3}x_2 + \frac{1}{3}x_3 + \frac{1}{2}w_1 + \frac{2}{3}w_2 + \frac{4}{9}w_3$$

Design variables $\underline{x} = \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix}$

Objective is to maximize $(x_1 + x_2 + x_3)$.

So $\min_{x_1, x_2, x_3} f(x_1, x_2, x_3) = -(x_1 + x_2 + x_3)$

s.t

$$T_A \leq W_1$$

$$T_B \leq W_1$$

$$T_C \leq W_2$$

$$T_D \leq W_2$$

$$T_E \leq W_3$$

$$T_F \leq W_3$$

$$\begin{cases} x_1 \geq 0 \\ x_2 \geq 0 \\ x_3 \geq 0 \end{cases}$$

non-negativity requirement.

NOTE that since the objective and constraints are linear functions (powers 0 or 1) of x_1, x_2, x_3 , this problem is a Linear Programming (LP) problem.

